T3 A_-2: full general UV Lagrangian and RGEs

Matchete/RGBeta symbolic output in mathematical notation

A_-2

Additional quartics: 5, additional bilinears: 1, extra Yukawa: y2Conj.
General matching and Wilson coefficients are not computed in this symbolic report.

Coupling and interaction notation

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C_{LLS_1S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$
R_W	$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$

General-interaction coupling index

Coupling	Field content / definition	Contraction / source

λ_{X_1}	$(S_2^\dagger)^4$	Contraction $I = 1$; exact tensor in registry `lambdaT3GeneralQ000004I1`
λ_{X_2}	$S_2\,(S_2^\dagger)^3$	Contraction $I = 1$; exact tensor in registry `lambdaT3GeneralQ000013I1`
$\lambda_{X_5}^{(1)}$	$S_1\,S_1^\dagger\,(S_2^\dagger)^2$	`del[SU2L @ S2, #3, #4]` `lambdaT3GeneralQ001102I1`
λ_{X_8}	$(H^\dagger)^2\,S_1^\dagger\,S_2^\dagger$	`delS2[SU2L, #4, #1, #2]` `lambdaT3GeneralQ020101I1`
$\lambda_{X_{11}}$	$H\,H^\dagger\,(S_2^\dagger)^2$	`(del[SU2L @ fund, #2, #1] * del[SU2L @ S2, #3, #4])` `lambdaT3GeneralQ110002I1`
$\mu_{\mathrm{T3GeneralS2Sq}}$	General scalar bilinear / mass parameter	`muT3GeneralS2Sq`
y_2^{conj}	Additional conjugate-scalar Yukawa	`y2Conj`

Each X_i labels one field-content class in the general-quartic registry; a superscript distinguishes different invariant contractions where present. The original RGBeta identifier and contraction are retained in the index. This is notation only, not a physical normalisation mapping.

Complete BSM UV Lagrangian

Matchete BSM Lagrangian, excluding the pure Standard Model contribution. $\mathcal{L}_{\rm BSM} =$ Coupling(lambdaT3ExtraM000002l1, {}, 0)CG(T3ExtraCGM000002l1, {Index(d$$1, T3BSMd3), Inde

+Coupling($\lambda_{x_\cdot}$, {}, 0)Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Bar(Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {}))Bar(Field(NewScalar2, Scalar, {Index(d$$3, T3BSMd3)}, {}))Bar(Field(NewScalar2, Scalar, {Index(d$$4, T3BSMd3)}, {}))CG(T3ExtraCGQ000004l1, {Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3), Index(d$$3, T3BSMd3), Index(d$$4, T3BSMd3)})

+Coupling(λ_{x_4}, {}, 0)Bar(Field(NewScalar1, Scalar, {}, {}))Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Bar(Field(H, Scalar, {Index(d$$1, SU2L(fund))}, {}))Bar(Field(H, Scalar, {Index(d$$2, SU2L(fund))}, {}))CG(T3ExtraCGQ020101l1, {Index(d$$1, SU2L(fund)), Index(d$$2, SU2L(fund)), Index(d$$1, T3BSMd3)})

+Coupling($\lambda_{x_{11}}$, {}, 0)Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Bar(Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {}))Field(H, Scalar, {Index(d$$2, SU2L(fund))}, {})Bar(Field(H, Scalar, {Index(d$$1, SU2L(fund))}, {}))CG(T3ExtraCGQ110002l1, {Bar(Index(d$$2, SU2L(fund))), Index(d$$1, SU2L(fund)), Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3)})

+Coupling($\lambda^{(1)}_{X_3}$, {}, 0)Field(NewScalar1, Scalar, {}, {})Bar(Field(NewScalar1, Scalar, {}, {}))CG(T3ExtraCGQ001102l1, {Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3)})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Bar(Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {}))

$$+\text{Coupling}(\text{MS1}, \{\}, 0)^2\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\})(-\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\})))$$

+Coupling(lambdaH1Inv1, {}, 0)Field(NewScalar1, Scalar, {}, {})Bar(Field(NewScalar1, Scalar, {}, {}))Field(H, Scalar, {Index(d$$2, SU2L(fund))}, {})CG(T3HiggsPortal1CGInv1, {Index(d$$1, SU2L(fund)), Bar(Index(d$$2, SU2L(fund)))})Bar(Field(H, Scalar, {Index(d$$1, SU2L(fund))}, {}))

+Coupling(lambdaT3, {}, 0)Field(NewScalar1, Scalar, {}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Field(H, Scalar, {Index(d$$1, SU2L(fund))}, {})Field(H, Scalar, {Index(d$$2, SU2L(fund))}, {})CG(T3MixCG, {Bar(Index(d$$1, SU2L(fund))), Bar(Index(d$$2, SU2L(fund))), Index(d$$1, T3BSMd3)})

$$+\frac{1}{2}\text{Coupling}(\text{lambdaS1Inv1}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\})^2\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\}))^2$$
$$+\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\}))$$

+Bar(Coupling(lambdaT3, {}, 0))Bar(Field(NewScalar1, Scalar, {}, {}))Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {})Bar(Field(H, Scalar, {Index(d$$1, SU2L(fund))}, {}))Bar(Field(H, Scalar, {Index(d$$2, SU2L(fund))}, {}))CG(Bar(T3MixCG), {Index(d$$1, SU2L(fund)), Index(d$$2, SU2L(fund)), Index(d$$1, T3BSMd3)})

$$+\text{Coupling}(\text{MS2}, \{\}, 0)^2\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})(-\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\})))$$

+Bar(Coupling(λ_{x_4}, {}, 0))Field(NewScalar1, Scalar, {}, {})Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {})Field(H, Scalar, {Index(d$$1, SU2L(fund))}, {})Field(H, Scalar, {Index(d$$2, SU2L(fund))}, {})CG(Bar(T3ExtraCGQ020101l1), {Bar(Index(d$$1, SU2L(fund))), Bar(Index(d$$2, SU2L(fund))), Index(d$$1, T3BSMd3)})

$$+\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3})\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\}))$$

+$\frac{1}{2}$Coupling(lambdaH2Inv1, {}, 0)Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Field(H, Scalar, {Index(d$$2, SU2L(fund))}, {})Bar(Field(H, Scalar, {Index(d$$1, SU2L(fund))}, {}))CG(T3HiggsPortal2CGInv1, {Index(d$$1, SU2L(fund)), Bar(Index(d$$2, SU2L(fund))), Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3)})

+$\frac{1}{2}$Coupling(lambdaH2Inv1, {}, 0)Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Field(H, Scalar, {Index(d$$2, SU2L(fund))}, {})Bar(Field(H, Scalar, {Index(d$$1, SU2L(fund))}, {}))CG(T3HiggsPortal2CGInv1, {Index(d$$1, SU2L(fund)), Bar(Index(d$$2, SU2L(fund))), Index(d$$2, T3BSMd3), Index(d$$1, T3BSMd3)})

+$\frac{1}{2}$Coupling(lambdaH2Inv2, {}, 0)Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Field(H, Scalar, {Index(d$$2, SU2L(fund))}, {})Bar(Field(H, Scalar, {Index(d$$1, SU2L(fund))}, {}))CG(T3HiggsPortal2CGInv2, {Index(d$$1, SU2L(fund)), Bar(Index(d$$2, SU2L(fund))), Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3)})

+$\frac{1}{2}$Coupling(lambdaH2Inv2, {}, 0)Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Field(H, Scalar, {Index(d$$2, SU2L(fund))}, {})Bar(Field(H, Scalar, {Index(d$$1, SU2L(fund))}, {}))CG(T3HiggsPortal2CGInv2, {Index(d$$1, SU2L(fund)), Bar(Index(d$$2, SU2L(fund))), Index(d$$2, T3BSMd3), Index(d$$1, T3BSMd3)})

+Coupling(lambda12Inv1, {}, 0)Field(NewScalar1, Scalar, {}, {})Bar(Field(NewScalar1, Scalar, {}, {}))CG(T3CrossScalarCGInv1, {Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3)})Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))

+Bar(Coupling(lambdaT3ExtraM000002l1, {}, 0))CG(T3ExtraCGM000002l1, {Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3)})Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})

+Bar(Coupling($\lambda_{x_{11}}$, {}, 0))Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})Field(H, Scalar, {Index(d$$2, SU2L(fund))}, {})Bar(Field(H, Scalar, {Index(d$$1, SU2L(fund))}, {}))CG(T3ExtraCGQ110002l1, {Bar(Index(d$$2, SU2L(fund))), Index(d$$1, SU2L(fund)), Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3)})

+Bar(Coupling($\lambda^{(1)}_{X_3}$, {}, 0))Field(NewScalar1, Scalar, {}, {})Bar(Field(NewScalar1, Scalar, {}, {}))CG(T3ExtraCGQ001102l1, {Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3)})Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})

+Coupling(λ_{x_7}, {}, 0)Field(NewScalar2, Scalar, {Index(d$$4, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Bar(Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {}))Bar(Field(NewScalar2, Scalar, {Index(d$$3, T3BSMd3)}, {}))CG(T3ExtraCGQ000013l1, {Index(d$$4, T3BSMd3), Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3), Index(d$$3, T3BSMd3)})

+$\frac{1}{2}$Coupling(lambdaS2Inv1, {}, 0)Field(NewScalar2, Scalar, {Index(d$$3, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$4, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Bar(Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {}))CG(T3Scalar2SdfCGInv1, {Index(d$$1, T3BSMd3), Index(d$$3, T3BSMd3), Index(d$$2, T3BSMd3), Index(d$$4, T3BSMd3)})

+$\frac{1}{2}$Coupling(lambdaS2Inv2, {}, 0)Field(NewScalar2, Scalar, {Index(d$$3, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$4, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))Bar(Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {}))CG(T3Scalar2SdfCGInv2, {Index(d$$1, T3BSMd3), Index(d$$3, T3BSMd3), Index(d$$2, T3BSMd3), Index(d$$4, T3BSMd3)})

+Bar(Coupling(λ_{x_5}, {}, 0))Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$3, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$4, T3BSMd3)}, {})Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))CG(T3ExtraCGQ000013l1, {Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3), Index(d$$3, T3BSMd3), Index(d$$4, T3BSMd3)})

+Bar(Coupling(λ_{X_1}, {}, 0))Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$2, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$3, T3BSMd3)}, {})Field(NewScalar2, Scalar, {Index(d$$4, T3BSMd3)}, {})CG(T3ExtraCGQ000004l1, {Index(d$$1, T3BSMd3), Index(d$$2, T3BSMd3), Index(d$$3, T3BSMd3), Index(d$$4, T3BSMd3)})

$$-\text{Coupling}(\text{MF}, \{\}, 0)\text{NCM}(\text{Bar}(\text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\})), \text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\}))$$

+Coupling(y2, {Index(d$$1, Flavor), Index(d$$1, NFlavor)}, 0)Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))CG(T3Y2CG, {Index(d$$1, SU2L(fund)), Bar(Index(d$$1, T3BSMd2)), Index(d$$1, T3BSMd3)})NCM(Bar(Field(l, Fermion, {Index(d$$1, SU2L(fund)), Index(d$$1, Flavor)}, {})), DiracProduct(Proj(1)), Field(NewFermion, Fermion, {Index(d$$1, T3BSMd2), Index(d$$1, NFlavor)}, {}))

+Coupling($y_2^{\rm conj}$, {Index(d$$1, Flavor), Index(d$$1, NFlavor)}, 0)Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {})CG(T3Y2ConjCG, {Index(d$$1, SU2L(fund)), Bar(Index(d$$1, T3BSMd2)), Index(d$$1, T3BSMd3)})NCM(Bar(Field(l, Fermion, {Index(d$$1, SU2L(fund)), Index(d$$1, Flavor)}, {})), DiracProduct(Proj(1)), Field(NewFermion, Fermion, {Index(d$$1, T3BSMd2), Index(d$$1, NFlavor)}, {}))

+Field(NewScalar1, Scalar, {}, {})Coupling(y1, {Index(d$$1, Flavor), Index(d$$1, NFlavor)}, 0)CG(T3Y1CG, {Index(d$$1, SU2L(fund)), Index(d$$1, T3BSMd2)})NCM(Bar(Field(l, Fermion, {Index(d$$1, SU2L(fund)), Index(d$$1, Flavor)}, {})), DiracProduct(GammaCC, Proj(1)), Transp(Bar(Field(NewFermion, Fermion, {Index(d$$1, T3BSMd2), Index(d$$1, NFlavor)}, {}))))

$$+i\text{NCM}(\text{Bar}(\text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\})), \text{DiracProduct}(\text{GammaM}(\text{Index}(\text{d\$\$1}, \text{Lorentz}))), \text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\}))$$

+Bar(Coupling($y_2^{\rm conj}$, {Index(d$$1, Flavor), Index(d$$1, NFlavor)}, 0))Bar(Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {}))CG(Bar(T3Y2ConjCG), {Bar(Index(d$$1, SU2L(fund))), Index(d$$1, T3BSMd2), Index(d$$1, T3BSMd3)})NCM(Bar(Field(NewFermion, Fermion, {Index(d$$1, T3BSMd2), Index(d$$1, NFlavor)}, {})), DiracProduct(Proj(−1)), Field(l, Fermion, {Index(d$$1, SU2L(fund)), Index(d$$1, Flavor)}, {}))

+Field(NewScalar2, Scalar, {Index(d$$1, T3BSMd3)}, {})Bar(Coupling(y2, {Index(d$$1, Flavor), Index(d$$1, NFlavor)}, 0))CG(Bar(T3Y2CG), {Bar(Index(d$$1, SU2L(fund))), Index(d$$1, T3BSMd2), Index(d$$1, T3BSMd3)})NCM(Bar(Field(NewFermion, Fermion, {Index(d$$1, T3BSMd2), Index(d$$1, NFlavor)}, {})), DiracProduct(Proj(−1)), Field(l, Fermion, {Index(d$$1, SU2L(fund)), Index(d$$1, Flavor)}, {}))

+Bar(Field(NewScalar1, Scalar, {}, {}))Bar(Coupling(y1, {Index(d$$1, Flavor), Index(d$$1, NFlavor)}, 0))CG(Bar(T3Y1CG), {Bar(Index(d$$1, SU2L(fund))), Bar(Index(d$$1, T3BSMd2))})NCM(Transp(Field(l, Fermion, {Index(d$$1, SU2L(fund)), Index(d$$1, Flavor)}, {})), DiracProduct(GammaCC, Proj(−1)), Field(NewFermion, Fermion, {Index(d$$1, T3BSMd2), Index(d$$1, NFlavor)}, {}))

Interaction inventory

- `Yukawa1_GenericCG`
- `Yukawa2_GenericCG`
- `Yukawa2ConjugateScalar_GenericCG`
- `Scalar1SelfInv1`
- `Scalar2SelfInv1`
- `Scalar2SelfInv2`
- `HiggsPortal1Inv1`

- HiggsPortal2Inv1
- HiggsPortal2Inv2
- ScalarCrossPortalInv1
- T3ScalarMix_GenericCG
- T3ExtraQ_0_0_0_0_0_4_1
- T3ExtraQ_0_0_0_0_1_3_1
- T3ExtraQ_0_0_1_1_0_2_1
- T3ExtraQ_0_2_0_1_0_1_1
- T3ExtraQ_1_1_0_0_0_2_1
- T3ExtraM_0_0_0_0_0_2_1

One-loop renormalisable beta functions

$16\pi^2\, dX/d\ln\mu = \beta_X$

g_1

$$16\pi^2 \frac{dg_1}{d\ln\mu} = \frac{55{g_1}^3}{6}$$

g_2

$$16\pi^2 \frac{dg_2}{d\ln\mu} = -\frac{{g_2}^3}{2}$$

g_3

$$16\pi^2 \frac{dg_3}{d\ln\mu} = -7{g_3}^3$$

Y_u

$$\begin{aligned}
16\pi^2 \frac{dY_u}{d\ln\mu} = {} & 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_u)_{ij} \\
& + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_u)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_u)_{ij} \\
& - \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_u\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_u\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_u)_{ij} \\
& - 8{g_3}^2\,(Y_u)_{ij} \\
& - \frac{17}{12}{g_1}^2\,(Y_u)_{ij}
\end{aligned}$$

Y_d

$$\begin{aligned}
16\pi^2 \frac{dY_d}{d\ln\mu} = {} & \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij} \\
& + \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_d\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_d\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_d)_{ij} \\
& - 8{g_3}^2\,(Y_d)_{ij} \\
& - \frac{5}{12}{g_1}^2\,(Y_d)_{ij}
\end{aligned}$$

Y_e

$$\begin{aligned}16\pi^2 \frac{dY_e}{d\ln\mu} &= 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_e)_{ij} \\ &+ 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_e)_{ij} \\ &+ \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_e)_{ij} \\ &+ \frac{1}{4}(2\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + 3(\left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + \left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ Y_e\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})) \\ &- \frac{9}{4}{g_2}^2\,(Y_e)_{ij} \\ &- \frac{15}{4}{g_1}^2\,(Y_e)_{ij}\end{aligned}$$

y_1

$$\begin{aligned}16\pi^2 \frac{dy_1}{d\ln\mu} &= 2T_\nu^{(1)}\,(y_1)_{ij} \\ &+ \frac{1}{4}(3\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij} + 3\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ y_1\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij} + 4\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij}) \\ &- \frac{9}{2}{g_2}^2\,(y_1)_{ij} \\ &- \frac{3}{2}{g_1}^2\,(y_1)_{ij}\end{aligned}$$

y_2

$$\begin{aligned}16\pi^2 \frac{dy_2}{d\ln\mu} &= \mathrm{Tr}\Big(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\Big)\,(y_2)_{ij} \\ &+ 2\,\mathrm{Tr}\Big(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\Big)\left(y_2^{\mathrm{conj}}\right)_{ij} \\ &+ T_\nu^{(2)}\,(y_2)_{ij} \\ &+ \frac{1}{12}(6\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + 29\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ y_2^{\mathrm{conj}}\right)_{ij} + 29\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ y_2\right)_{ij} + 6\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 18\left(y_2\ (y_2)\right. \\ &- \frac{9}{2}{g_2}^2\,(y_2)_{ij} \\ &- \frac{3}{2}{g_1}^2\,(y_2)_{ij}\end{aligned}$$

M_F

$$\begin{aligned}16\pi^2 \frac{dM_F}{d\ln\mu} &= \frac{1}{4}(2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + 3\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij} + 3\left(M_F\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ y_2^{\mathrm{conj}}\right)_{ij}) \\ &- \frac{9}{2}{g_2}^2\,(M_F)_{ij} \\ &- \frac{3}{2}{g_1}^2\,(M_F)_{ij}\end{aligned}$$

m_1^2

$$\begin{aligned}16\pi^2 \frac{dm_1^2}{d\ln\mu} = & m_1^2(4T_\nu^{(1)} + 4\lambda_{S_1}^{(1)}) \\ & - 8\,\mathrm{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big) \\ & + 12\mu_{\mathrm{T3GeneralS2Sq}}\left(\lambda_{X_5}^{(1)}\right)^* \\ & + 12\lambda_{X_5}^{(1)}\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^* \\ & - 6{g_1}^2 m_1^2 \\ & + 6\lambda_{12}^{(1)} m_2^2\end{aligned}$$

m_2^2

$$\begin{aligned}16\pi^2 \frac{dm_2^2}{d\ln\mu} = & m_2^2(2T_\nu^{(2)} + 2\,\mathrm{Tr}\Big(y_2^{\mathrm{conj}}\ \Big(y_2^{\mathrm{conj}}\Big)^\dagger\Big) + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)}) \\ & + \mu_{\mathrm{T3GeneralS2Sq}}(4\,\mathrm{Tr}\Big(y_2^{\mathrm{conj}}\ (y_2)^\dagger\Big) + 20\,(\lambda_{X_2})^*) \\ & - 4\,\mathrm{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\Big) \\ & - 4\,\mathrm{Tr}\Big(M_F\ \Big(y_2^{\mathrm{conj}}\Big)^\dagger\ y_2^{\mathrm{conj}}\ (M_F)^\dagger\Big) \\ & + 4\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^* \mathrm{Tr}\Big(y_2\ \Big(y_2^{\mathrm{conj}}\Big)^\dagger\Big) \\ & + 20\lambda_{X_2}\left(\mu_{\mathrm{T3GeneralS2Sq}}\right)^* \\ & - 12{g_2}^2 m_2^2 \\ & + 2\lambda_{12}^{(1)} m_1^2\end{aligned}$$

λ_1

$$\begin{aligned}16\pi^2 \frac{d\lambda_1}{d\ln\mu} = & \lambda_1(12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\ & - 12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big) \\ & - 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\ & - 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big) \\ & + 4\lambda_5\left(\lambda_5\right)^* \\ & + 4\lambda_{X_8}\left(\lambda_{X_8}\right)^* \\ & + 24\lambda_{X_{11}}\left(\lambda_{X_{11}}\right)^* \\ & + \frac{9{g_2}^4}{4} \\ & + {g_1}^2\left(\frac{3{g_2}^2}{2} - 3\lambda_1\right) \\ & - 9{g_2}^2\lambda_1 \\ & + \frac{3{g_1}^4}{4} \\ & + 12{\lambda_1}^2 \\ & + 2{\lambda_{HS_1}^{(1)}}^2 \\ & + 6{\lambda_{HS_2}^{(1)}}^2\end{aligned}$$

$$+ 3{\lambda_{HS_2}^{(A)}}^2$$

$\lambda_{S_1}^{(1)}$

$$\begin{aligned}
16\pi^2 \frac{d\lambda_{S_1}^{(1)}}{d\ln\mu} = & 8\lambda_{S_1}^{(1)} T_\nu^{(1)} \\
& - 8\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big) \\
& + 24\lambda_{X_5}^{(1)} \left(\lambda_{X_5}^{(1)}\right)^* \\
& + 12{g_1}^4 \\
& - 12{g_1}^2 \lambda_{S_1}^{(1)} \\
& + 6{\lambda_{12}^{(1)}}^2 \\
& + 4{\lambda_{HS_1}^{(1)}}^2 \\
& + 10{\lambda_{S_1}^{(1)}}^2
\end{aligned}$$

$\lambda_{S_2}^{(1)}$

$$\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(1)}}{d\ln\mu} = & \lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 8\lambda_{S_2}^{(A)}) \\
& + \frac{32}{3}\left(\lambda_{X_2}\right)^* \mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) \\
& + \lambda_{X_2}\left(\frac{32}{3}\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ (y_2)^\dagger\right) + \frac{352\left(\lambda_{X_2}\right)^*}{3}\right) \\
& - \frac{248}{45}\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) \\
& - \frac{248}{45}\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ y_2^{\mathrm{conj}}\ (y_2)^\dagger\right) \\
& - \frac{8}{3}\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big) \\
& - \frac{8}{3}\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) \\
& + \frac{832}{3}\lambda_{X_1}\left(\lambda_{X_1}\right)^* \\
& + \frac{8}{3}\lambda_{X_5}^{(1)}\left(\lambda_{X_5}^{(1)}\right)^* \\
& + \frac{16}{3}\lambda_{X_{11}}\left(\lambda_{X_{11}}\right)^* \\
& + 22{g_2}^4 \\
& - 24{g_2}^2\lambda_{S_2}^{(1)} \\
& + 2{\lambda_{12}^{(1)}}^2 \\
& + 4{\lambda_{HS_2}^{(1)}}^2 \\
& + 14{\lambda_{S_2}^{(1)}}^2 \\
& + 16{\lambda_{S_2}^{(A)}}^2
\end{aligned}$$

$\lambda_{HS_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_1}^{(1)}}{d\ln\mu} = {} & \lambda_{HS_1}^{(1)}(4T_\nu^{(1)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
& - 4\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\
& + 12\lambda_5\,(\lambda_5)^* \\
& + 12\lambda_{X_{11}}\left(\lambda_{X_5}^{(1)}\right)^* \\
& + 12\lambda_{X_5}^{(1)}\,(\lambda_{X_{11}})^* \\
& + 12\lambda_{X_8}\,(\lambda_{X_8})^* \\
& - \frac{9{g_2}^2\lambda_{HS_1}^{(1)}}{2} \\
& + 3{g_1}^4 \\
& - \frac{15{g_1}^2\lambda_{HS_1}^{(1)}}{2} \\
& + 6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)} \\
& + 6\lambda_1\lambda_{HS_1}^{(1)} \\
& + 4{\lambda_{HS_1}^{(1)}}^2 \\
& + 4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}
\end{aligned}
$$

$\lambda_{HS_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_2}^{(1)}}{d\ln\mu} = {} & \lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 4\lambda_{S_2}^{(A)}) \\
& + \lambda_{X_{11}}(4\,\mathrm{Tr}\Big(y_2^{\mathrm{conj}}\ (y_2)^\dagger\Big) + 4(5\,(\lambda_{X_2})^* + 4\,(\lambda_{X_{11}})^*)) \\
& + 4\,(\lambda_{X_{11}})^*\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) \\
& - 2\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\
& - 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ Y_e\ (Y_e)^\dagger\right) \\
& + 4\lambda_5\,(\lambda_5)^* \\
& + 20\lambda_{X_2}\,(\lambda_{X_{11}})^* \\
& + 4\lambda_{X_8}\,(\lambda_{X_8})^* \\
& + 6{g_2}^4 \\
& - \frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2} \\
& - \frac{3{g_1}^2\lambda_{HS_2}^{(1)}}{2} \\
& + 2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)} \\
& + 6\lambda_1\lambda_{HS_2}^{(1)} \\
& + 4{\lambda_{HS_2}^{(1)}}^2 \\
& + 8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
& + 6{\lambda_{HS_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{12}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{12}^{(1)}}{d\ln\mu} = {} & \lambda_{12}^{(1)}(4T_\nu^{(1)} + 2T_\nu^{(2)} + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 4\lambda_{S_2}^{(A)}) \\
& + \lambda_{X_5}^{(1)}(4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ (y_2)^\dagger\right) + 4(5\,(\lambda_{X_2})^* + 4\left(\lambda_{X_5}^{(1)}\right)^*)) \\
& + 4\left(\lambda_{X_5}^{(1)}\right)^* \mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) \\
& - 4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right) \\
& - 4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) \\
& + 4\lambda_5\,(\lambda_5)^* \\
& + 20\lambda_{X_2}\left(\lambda_{X_5}^{(1)}\right)^* \\
& + 4\lambda_{X_8}\,(\lambda_{X_8})^* \\
& - 12{g_2}^2\lambda_{12}^{(1)} \\
& - 6{g_1}^2\lambda_{12}^{(1)} \\
& + 4{\lambda_{12}^{(1)}}^2 \\
& + 4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)} \\
& + 8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)} \\
& + 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}
\end{aligned}
$$

λ_5

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_5}{d\ln\mu} = {} & \lambda_5(2T_\nu^{(1)} + T_\nu^{(2)} + \mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda_{HS_2}^{(A)}) \\
& + 2\,(\lambda_{X_8})^*\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) \\
& + 4\lambda_{X_5}^{(1)}\,(\lambda_{X_8})^* \\
& + 8\lambda_{X_{11}}\,(\lambda_{X_8})^* \\
& - \frac{21{g_2}^2\lambda_5}{2} \\
& - \frac{9{g_1}^2\lambda_5}{2} \\
& + 2\lambda_{12}^{(1)}\lambda_5 \\
& + 2\lambda_1\lambda_5 \\
& + 4\lambda_{HS_1}^{(1)}\lambda_5 \\
& + 4\lambda_{HS_2}^{(1)}\lambda_5
\end{aligned}
$$

$\lambda_{HS_2}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_2}^{(A)}}{d\ln\mu} = {} & \lambda_{HS_2}^{(A)}(2T_\nu^{(2)} + 2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda_{S_2}^{(A)}) \\
& + \frac{4}{3}\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right) \\
& - \frac{4}{3}\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ Y_e\ (Y_e)^\dagger\right)
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{8}{3}\lambda_5 \left(\lambda_5\right)^* \\
&- \frac{8}{3}\lambda_{X_8} \left(\lambda_{X_8}\right)^* \\
&- \frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2} \\
&- \frac{3{g_1}^2\lambda_{HS_2}^{(A)}}{2} \\
&+ 2\lambda_1\lambda_{HS_2}^{(A)} \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}
\end{aligned}
$$

$\lambda_{S_2}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(A)}}{d\ln\mu} = &\lambda_{S_2}^{(A)}(4T_\nu^{(2)} + 4\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)) \\
&- \frac{4}{3}\left(\lambda_{X_2}\right)^* \mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) \\
&+ \lambda_{X_2}\left(-\frac{4}{3}\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2\right)^\dagger\right) - \frac{44\left(\lambda_{X_2}\right)^*}{3}\right) \\
&+ \frac{76}{45}\,\mathrm{Tr}\left(y_2\ \left(y_2\right)^\dagger\ y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) \\
&+ \frac{76}{45}\,\mathrm{Tr}\left(y_2\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ y_2^{\mathrm{conj}}\ \left(y_2\right)^\dagger\right) \\
&- \frac{2}{3}\,\mathrm{Tr}\left(y_2\ \left(y_2\right)^\dagger\ y_2\ \left(y_2\right)^\dagger\right) \\
&- \frac{2}{3}\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\ y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right) \\
&- \frac{224}{3}\lambda_{X_1}\left(\lambda_{X_1}\right)^* \\
&- \frac{4}{3}\lambda_{X_5}^{(1)}\left(\lambda_{X_5}^{(1)}\right)^* \\
&- \frac{8}{3}\lambda_{X_{11}}\left(\lambda_{X_{11}}\right)^* \\
&- 2{g_2}^4 \\
&- 24{g_2}^2\lambda_{S_2}^{(A)} \\
&+ 3{\lambda_{HS_2}^{(A)}}^2 \\
&+ 12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\
&+ 20{\lambda_{S_2}^{(A)}}^2
\end{aligned}
$$

y_2^{conj}

$$
\begin{aligned}
16\pi^2 \frac{dy_2^{\mathrm{conj}}}{d\ln\mu} = &\,2\,\mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2\right)^\dagger\right)\left(y_2\right)_{ij} \\
&+ T_\nu^{(2)}\left(y_2^{\mathrm{conj}}\right)_{ij} \\
&+ \mathrm{Tr}\left(y_2^{\mathrm{conj}}\ \left(y_2^{\mathrm{conj}}\right)^\dagger\right)\left(y_2^{\mathrm{conj}}\right)_{ij}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{1}{12}(6 \left(y_1 \ (y_1)^\dagger \ y_2^{\text{conj}}\right)_{ij} + 29 \left(y_2 \ (y_2)^\dagger \ y_2^{\text{conj}}\right)_{ij} + 29 \left(y_2^{\text{conj}} \ (y_2)^\dagger \ y_2\right)_{ij} + 6 \left(Y_e \ (Y_e)^\dagger \ y_2^{\text{conj}}\right)_{ij} + 18 \left(y_2^{\text{conj}} \ \left(y_2^{\text{co}}\right.\right.\\
&- \frac{9}{2} {g_2}^2 \left(y_2^{\text{conj}}\right)_{ij}\\
&- \frac{3}{2} {g_1}^2 \left(y_2^{\text{conj}}\right)_{ij}
\end{aligned}
$$

λ_{X_1}

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{X_1}}{d\ln\mu} = &\lambda_{X_1}(4T_\nu^{(2)} + 4 \operatorname{Tr}\left(y_2^{\text{conj}} \ \left(y_2^{\text{conj}}\right)^\dagger\right))\\
&+ 2\lambda_{X_2} \operatorname{Tr}\left(y_2 \ \left(y_2^{\text{conj}}\right)^\dagger\right)\\
&- \frac{23}{15} \operatorname{Tr}\left(y_2 \ \left(y_2^{\text{conj}}\right)^\dagger \ y_2 \ \left(y_2^{\text{conj}}\right)^\dagger\right)\\
&- 24 {g_2}^2 \lambda_{X_1}\\
&+ 12\lambda_{S_2}^{(1)} \lambda_{X_1}\\
&- 8\lambda_{S_2}^{(A)} \lambda_{X_1}\\
&+ 11 {\lambda_{X_2}}^2\\
&+ {\lambda_{X_5}^{(1)}}^2\\
&+ 2 {\lambda_{X_{11}}}^2
\end{aligned}
$$

λ_{X_2}

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{X_2}}{d\ln\mu} = &\lambda_{S_2}^{(1)}(2 \operatorname{Tr}\left(y_2 \ \left(y_2^{\text{conj}}\right)^\dagger\right) + 22\lambda_{X_2})\\
&+ \lambda_{X_1}(8 \operatorname{Tr}\left(y_2^{\text{conj}} \ (y_2)^\dagger\right) + 88 \left(\lambda_{X_2}\right)^*)\\
&+ \lambda_{X_2}(4T_\nu^{(2)} + 4 \operatorname{Tr}\left(y_2^{\text{conj}} \ \left(y_2^{\text{conj}}\right)^\dagger\right))\\
&- \frac{46}{15} \operatorname{Tr}\left(y_2 \ (y_2)^\dagger \ y_2 \ \left(y_2^{\text{conj}}\right)^\dagger\right)\\
&- \frac{46}{15} \operatorname{Tr}\left(y_2 \ \left(y_2^{\text{conj}}\right)^\dagger \ y_2^{\text{conj}} \ \left(y_2^{\text{conj}}\right)^\dagger\right)\\
&- 24 {g_2}^2 \lambda_{X_2}\\
&+ 2\lambda_{12}^{(1)} \lambda_{X_5}^{(1)}\\
&+ 4\lambda_{HS_2}^{(1)} \lambda_{X_{11}}
\end{aligned}
$$

$\lambda_{X_5}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{X_5}^{(1)}}{d\ln\mu} = &\lambda_{12}^{(1)}(2 \operatorname{Tr}\left(y_2 \ \left(y_2^{\text{conj}}\right)^\dagger\right) + 10\lambda_{X_2} + 8\lambda_{X_5}^{(1)})\\
&+ \lambda_{X_5}^{(1)}(4T_\nu^{(1)} + 2T_\nu^{(2)} + 2 \operatorname{Tr}\left(y_2^{\text{conj}} \ \left(y_2^{\text{conj}}\right)^\dagger\right))\\
&- 4 \operatorname{Tr}\left(y_1 \ (y_1)^\dagger \ y_2 \ \left(y_2^{\text{conj}}\right)^\dagger\right)\\
&+ 40\lambda_{X_1} \left(\lambda_{X_5}^{(1)}\right)^*
\end{aligned}
$$

$$
\begin{aligned}
&- 12{g_2}^2\lambda_{X_5}^{(1)} \\
&- 6{g_1}^2\lambda_{X_5}^{(1)} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{X_{11}} \\
&+ 4\lambda_{S_1}^{(1)}\lambda_{X_5}^{(1)} \\
&+ 2\lambda_{S_2}^{(1)}\lambda_{X_5}^{(1)} \\
&- 4\lambda_{S_2}^{(A)}\lambda_{X_5}^{(1)} \\
&+ 4\lambda_5\lambda_{X_8}
\end{aligned}
$$

λ_{X_8}

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_{X_8}}{d\ln\mu} = {} & 2\left(\lambda_5\right)^*\operatorname{Tr}\left(y_2\,\left(y_2^{\text{conj}}\right)^\dagger\right) \\
& + \lambda_{X_8}(2T_\nu^{(1)} + T_\nu^{(2)} + \operatorname{Tr}\left(y_2^{\text{conj}}\,\left(y_2^{\text{conj}}\right)^\dagger\right) + 6\operatorname{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right) + 2\operatorname{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right) + 6\operatorname{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)) \\
& + 4\lambda_{X_5}^{(1)}\left(\lambda_5\right)^* \\
& + 8\lambda_{X_{11}}\left(\lambda_5\right)^* \\
& - \frac{21{g_2}^2\lambda_{X_8}}{2} \\
& - \frac{9{g_1}^2\lambda_{X_8}}{2} \\
& + 2\lambda_{12}^{(1)}\lambda_{X_8} \\
& + 2\lambda_1\lambda_{X_8} \\
& + 4\lambda_{HS_1}^{(1)}\lambda_{X_8} \\
& + 4\lambda_{HS_2}^{(1)}\lambda_{X_8} \\
& - 4\lambda_{HS_2}^{(A)}\lambda_{X_8}
\end{aligned}
$$

$\lambda_{X_{11}}$

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_{X_{11}}}{d\ln\mu} = {} & \lambda_{HS_2}^{(1)}(2\operatorname{Tr}\left(y_2\,\left(y_2^{\text{conj}}\right)^\dagger\right) + 10\lambda_{X_2} + 8\lambda_{X_{11}}) \\
& + \lambda_{X_{11}}(2T_\nu^{(2)} + 2\operatorname{Tr}\left(y_2^{\text{conj}}\,\left(y_2^{\text{conj}}\right)^\dagger\right) + 6\operatorname{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right) + 2\operatorname{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right) + 6\operatorname{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)) \\
& - 2\operatorname{Tr}\left(y_2\,\left(y_2^{\text{conj}}\right)^\dagger\,Y_e\,\left(Y_e\right)^\dagger\right) \\
& + 40\lambda_{X_1}\left(\lambda_{X_{11}}\right)^* \\
& - \frac{33{g_2}^2\lambda_{X_{11}}}{2} \\
& - \frac{3{g_1}^2\lambda_{X_{11}}}{2} \\
& + 6\lambda_1\lambda_{X_{11}} \\
& + 2\lambda_{HS_1}^{(1)}\lambda_{X_5}^{(1)} \\
& + 2\lambda_{S_2}^{(1)}\lambda_{X_{11}} \\
& - 4\lambda_{S_2}^{(A)}\lambda_{X_{11}} \\
& + 4\lambda_5\lambda_{X_8}
\end{aligned}
$$

$\mu_{\text{T3GeneralS2Sq}}$

$$
\begin{aligned}
16\pi^2 \frac{d\mu_{\text{T3GeneralS2Sq}}}{d\ln\mu} &= m_2^2(2\,\text{Tr}\left(y_2\ \left(y_2^{\text{conj}}\right)^\dagger\right) + 10\lambda_{X_2}) \\
&\quad - 4\,\text{Tr}\left(M_F\ \left(y_2^{\text{conj}}\right)^\dagger\ y_2\ (M_F)^\dagger\right) \\
&\quad + \mu_{\text{T3GeneralS2Sq}}(2T_\nu^{(2)} + 2\,\text{Tr}\left(y_2^{\text{conj}}\ \left(y_2^{\text{conj}}\right)^\dagger\right)) \\
&\quad + 40\lambda_{X_1}\left(\mu_{\text{T3GeneralS2Sq}}\right)^* \\
&\quad - 12{g_2}^2\mu_{\text{T3GeneralS2Sq}} \\
&\quad + 2\lambda_{S_2}^{(1)}\mu_{\text{T3GeneralS2Sq}} \\
&\quad - 4\lambda_{S_2}^{(A)}\mu_{\text{T3GeneralS2Sq}} \\
&\quad + 2\lambda_{X_5}^{(1)}m_1^2
\end{aligned}
$$